

Supervised Non-Generative Reranking for Fixed-Candidate MS/MS Molecule Retrieval

Abstract. Retrieving molecular structures from tandem mass spectra is difficult because mass- and formula-constrained candidate pools contain chemically similar hard negatives. We study whether supervised, non-generative second-stage reranking can improve a fixed candidate list without generating new structures. SpecEmbedding-Rerank first aligns a peak-sequence spectrum encoder with a molecular graph encoder and reuses saved embeddings for the supplied pool of at most 256 candidates. It then performs pointwise coarse scoring followed by a spectrum-conditioned relative candidate module on the coarse top 40. The relative preference is constructed as an antisymmetric pair difference and is equivariant to candidate-list permutation, subject to coarse-selection ties. No retrieval-rank feature is used. We compare the relative model with a near-parameter-matched no-rank pointwise control. On MassSpecGym mass- and formula-conditioned candidate pools, both supervised rerankers improve the fixed cross-modal retriever in a saved-embedding top-40 evaluation. These runs use up to 20,000 training queries and training-only positive forcing; they do not validate full-training, no-forcing use of the 256-candidate configuration. The relative-pointwise ordering depends on the pool and metric. We therefore present the relative branch as a formally defined candidate- interaction mechanism and study its empirical boundary, rather than claiming a universal relation-module or external-baseline advantage. The method is closed-set and non-generative: it improves the ordering of supplied candidates but cannot recover a molecule absent from that pool.

Keywords: Data mining for bioinformatics · Tandem mass spectrometry · Molecule retrieval · Learning to rank · Multimodal representation learning

1 Introduction

MS/MS is widely used to identify small molecules in metabolomics, natural-product discovery, and related biomedical applications. An observed spectrum is only a partial, condition-dependent view of a molecular structure: different molecules can yield similar fragments, while the same molecule can produce different spectra across instruments, adducts, and collision energies. Consequently, structure annotation is naturally a large-scale data mining problem in which a query spectrum must retrieve and rank a correct structure from a chemically constrained candidate pool.

MassSpecGym provides structure-disjoint data splits and standardized mass- and formula-conditioned candidate pools for this problem [3]. Recent methods

mainly address the cross-modal gap between spectra and molecules. JESTR learns a spectrum-molecule joint space with contrastive multiview coding and ranks candidates directly by cosine similarity [12,16]. GLMR instead retrieves contextual candidates, generates a refined molecule with a conditional language model, and reranks candidates by molecule-molecule similarity [20].

Both approaches leave an important data mining question open. Candidate-wise cosine scoring does not explicitly train a second-stage objective over the retrieved set, while generative retrieval adds an autoregressive decoder and makes ranking depend on the quality and validity of a generated structure. We therefore ask two related questions: can a supervised second-stage ranker separate hard candidates, and does spectrum-conditioned relative comparison add value beyond a near-parameter-matched pointwise scorer? The first question tests the value of learning to reorder a fixed list; the second tests whether candidate interaction is needed for that gain. Within each paired comparison, the candidate pool and first-stage representations are held fixed so that the comparison concerns the second-stage ranking mechanism; the cross-alignment matrix separately measures sensitivity to different saved first-stage checkpoints.

Building on the peak-sequence spectrum architecture introduced by SpecEmbedding [18], we construct SpecEmbedding-Rerank, a two-stage retrieval and reranking framework. The first stage trains that spectrum tower from scratch and aligns it with a GINE molecular graph encoder. The second stage scores the full supplied pool with an absolute branch, then refines only a coarse top- K_r subset with spectrum-conditioned candidate relations. Pair preferences are formed by subtracting the score of the reversed ordered pair, which gives antisymmetry without a rank embedding. The default pipeline uses naturally present positives; the reported top-40 matrix instead reuses historical training caches with positive forcing. A pointwise model with the same absolute branch provides the primary capacity-controlled comparison, while the earlier candidate-set Transformer is kept only as a historical reference.

Our contributions are:

- We formulate a direct, closed-set, non-generative second-stage learning-to-rank framework for hard candidates, using full-pool coarse scoring followed by spectrum-conditioned relative candidate comparison.
- We define a rank-free relative branch with antisymmetric pair preferences and candidate-list permutation equivariance, together with a near-parameter-matched pointwise control.
- We systematically examine the framework on MassSpecGym mass- and formula-conditioned pools, three saved alignment checkpoints, and a 30-run relative-module ablation matrix: supervised reranking improves the base retriever, while the evidence does not establish an independent advantage of relative interactions over pointwise scoring.

AI-assistance disclosure. OpenAI Codex assisted throughout all sections with drafting and language revision, and during the project with code editing, experiment orchestration, and consistency audits. Reported observations and numerical results came from the stated software pipeline, not model generation. The

authors reviewed all AI-assisted text and code, verified the technical claims, citations, saved artifacts, and reported results, and assume full responsibility for the manuscript’s accuracy, completeness, and originality.

2 Related Work

2.1 Annotation Paradigms

Computational MS/MS annotation differs in retrieved objects and ranking evidence. Candidate-score correction predates our work: MetFusion combines in-silico fragmentation with spectral-library evidence [8]; MS2Query reranks its top 2,000 MS2DeepScore matches with a five-feature random forest [11]; and LC-MS2Struct combines an MS/MS scorer with retention order to jointly rank candidates from one LC analysis [1]. We instead rank structures in query-specific MassSpecGym pools from frozen spectrum–molecule representations, without spectral-library-neighbor or retention-time evidence [3].

Within structure-database retrieval, CSI:FingerID and MIST predict molecular fingerprints from spectra [7,9]. CMSSP and JESTR learn joint spectrum–molecule spaces; MVP extends contrastive alignment across molecular graphs, fingerprints, individual spectra, and consensus spectra [5,12,6]. These methods order candidates by predicted-fingerprint scores or cross-modal similarity. SpecEmbedding instead studies spectrum–spectrum representation learning [18]; we reuse only its peak-sequence architecture, train it from scratch against molecular graphs, and learn a complementary second-stage order within a retrieved hard-candidate list.

Other paradigms use different intermediate objects. MassFormer predicts tandem spectra from molecular graphs [19], while MSNovelist generates structures from MS/MS-derived fingerprints [15]. GLMR conditions its molecular decoder on retrieved candidates and reranks them by similarity to the generated molecule [20]. Our closed-set method does not decode a structure or return a molecule outside the first-stage list; it directly optimizes the scores of available candidates.

2.2 Learning to Rank Candidate Sets

Learning-to-rank objectives define supervision over preferences or complete lists rather than treating similarity sorting as a fixed post-processing step. RankNet learns pairwise preferences, whereas ListNet defines a probability model over a ranked list [2,4]. Ranking functions can also exploit relations among competing items. Set Transformer provides attention blocks for set-structured inputs [13], while SetRank uses self-attention to model cross-document interactions [14]. We apply these principles to cached cross-modal embeddings with explicit spectrum–candidate pair features and first-stage score and rank priors. Table 1 positions the resulting query-specific, non-generative second stage among selected related MassSpecGym approaches. The contribution is neither reranking nor self-attention itself; residual scoring and candidate interaction are evaluated design choices rather than assumed sources of gain.

Table 1. Mechanistic comparison with selected related MassSpecGym approaches.

Model	Ranking signal	Candidate interaction	Generates	Training stages
JESTR	Spectrum-candidate cosine	None at inference	No	InfoNCE; optional late-stage candidate regularization
GLMR	Generated-candidate cosine	Retrieved candidates condition the generator	Yes	Dual-path InfoNCE plus generation
Ours	Residual score from pair features and retrieval priors	Spectrum-conditioned antisymmetric pair interaction	No	Cross-modal InfoNCE plus listwise/pairwise reranking

3 Problem Formulation

Let s be an MS/MS spectrum, m^+ its correct molecule, and $\mathcal{D}_s$ a query-specific mass- or formula-conditioned candidate pool. A base retriever assigns

$$b(s, m) = \text{sim}(f_s(s), f_m(m)) \quad (1)$$

to every $m \in \mathcal{D}_s$ and returns the top- K list $\mathcal{C}_s = (m_1, \dots, m_K)$. The reranker learns scores $g(s, m_i, \mathcal{C}_s)$ such that m^+ is ordered as high as possible. Because it cannot return a molecule outside $\mathcal{C}_s$, end-to-end Recall@ k is upper-bounded by

$$U_K = \frac{1}{N} \sum_{n=1}^N \mathbb{I}[m_n^+ \in \mathcal{C}_{s_n}]. \quad (2)$$

Define the within-list top- k accuracy conditional on first-stage coverage as

$$A_{k|K} = \frac{\sum_{n=1}^N \mathbb{I}[m_n^+ \in \mathcal{C}_{s_n}, \text{rank}_g(m_n^+) \leq k]}{\sum_{n=1}^N \mathbb{I}[m_n^+ \in \mathcal{C}_{s_n}]}, \quad \text{Recall@}k = U_K A_{k|K} \leq U_K. \quad (3)$$

A reranker applied to a fixed cache can therefore change only $A_{k|K}$, not U_K . We report metrics over all queries, assigning zero credit when the positive is absent rather than conditioning the primary metric on successful first-stage retrieval.

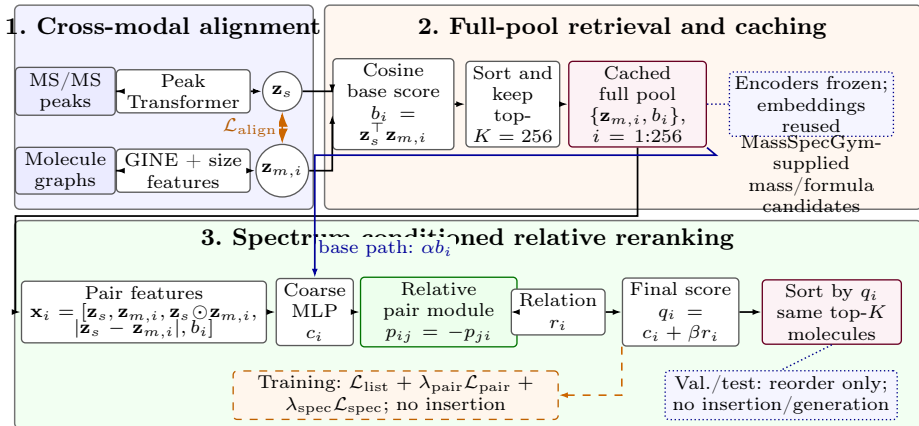


Fig. 1. Default top-256 configuration of SpecEmbedding-Rerank. Panel 1 aligns the spectrum Transformer and molecular GINE. Panel 2 uses their frozen, cached embeddings to produce the full candidate pool. Panel 3 applies pointwise coarse scoring followed by spectrum-conditioned relative candidate preferences on the coarse top-40; the legacy Transformer is a historical reference. Dashed arrows denote training-only operations. This default does not force positives or generate molecules; the reported matrix uses historical top-40 training caches with positive forcing.

4 Method

4.1 Overview

The framework has three steps: cross-modal alignment, full-pool retrieval, and supervised coarse-to-fine relative reranking. Spectrum and molecule encoders are frozen while the reranker is trained, so their embeddings can be computed once and reused.

4.2 Cross-Modal Base Retriever

The spectrum encoder follows SpecEmbedding’s peak-sequence architecture [18], which processes peak m/z values and intensities with sinusoidal m/z features and a Transformer. In our implementation, each spectrum has at most 100 tokens: a precursor- m/z token with intensity 2, followed by up to 99 fragments selected by intensity. Selected fragments retain their source-array order, and their intensities are normalized by the largest selected fragment intensity. Only these m/z -intensity tokens enter the spectrum encoder; adduct, instrument, and collision-energy metadata are not explicit inputs. We initialize this tower from scratch and train it against molecules rather than spectra. The molecular encoder is a GINE network over atom and bond features [10]. Mean-pooled graph features are augmented with explicit graph-size features. Two projection heads map the encoders to a common d -dimensional space:

$$\mathbf{z}_s = \frac{p_s(f_s(s))}{\|p_s(f_s(s))\|_2}, \quad \mathbf{z}_m = \frac{p_m(f_m(m))}{\|p_m(f_m(m))\|_2}. \quad (4)$$

We use a bidirectional cross-modal contrastive loss that permits multiple positives. Spectra and molecules sharing a molecular label form positives, which avoids treating replicate spectra as false negatives. For a minibatch of size B , let $P(i)$ be the indices sharing the label of item i . With a learned inverse-temperature scale a , the base logits are

$$\ell_{ij} = a \mathbf{z}_{s_i}^\top \mathbf{z}_{m_j}. \quad (5)$$

The spectrum-to-molecule term is

$$\mathcal{L}_{s \rightarrow m} = -\frac{1}{B} \sum_{i=1}^B \frac{1}{|P(i)|} \sum_{j \in P(i)} \log \frac{\exp(\ell_{ij})}{\sum_{k=1}^B \exp(\ell_{ik})}, \quad (6)$$

and the molecule-to-spectrum term applies the same expression to $\ell^\top$. The alignment loss is their mean. The implementation parameterizes $a = \exp(u)$ and caps it at 100. At inference, the normalized dot product $b_i = \mathbf{z}_s^\top \mathbf{z}_{m_i}$ is the base score.

4.3 Candidate Construction and Training Protocol

MassSpecGym supplies target-molecule-keyed candidate lists; each query reuses the list keyed by its target molecule. The base retriever scores the supplied pool of at most 256 molecules. The default coarse-to-fine configuration retains this pool for pointwise scoring and applies the relation module to the coarse top $K_r = 40$. The reported main and ablation matrices instead use saved base top-40 lists, so their relation branch operates on at most 40 candidates.

The default pipeline does not insert positives: unlabeled queries contribute to coverage statistics but not the training loss. In the reported top-40 runs, however, a missing training positive is inserted by replacing the last candidate. Validation and test never force positives; missing targets receive zero metric credit. These historical training caches must be distinguished from the default no-forcing configuration and the separate top-256 pilot. During training, candidate embeddings, scores and labels are shuffled together. Neither the relative model nor its no-rank control uses a retrieval-rank feature; the relative forward path ignores the cache-compatible `base_ranks` field.

4.4 Spectrum-Conditioned Relative Candidate Reranker

For candidate m_i we retain an absolute compatibility vector, but omit the retrieval rank:

$$\mathbf{x}_i = [\mathbf{z}_s, \mathbf{z}_{m_i}, \mathbf{z}_s \odot \mathbf{z}_{m_i}, |\mathbf{z}_s - \mathbf{z}_{m_i}|, b_i]. \quad (7)$$

An absolute MLP produces a coarse score $c_i = f_{\text{abs}}(\mathbf{x}_i) + \alpha b_i$. For the coarse top- K_r candidates, let $u_i = W_m \mathbf{z}_{m_i}$ and $v = W_s \mathbf{z}_s$. The relation vector ρ_{ij} contains the spectrum condition, the signed difference $u_i - u_j$, its absolute value, the base-score difference $b_i - b_j$, and the cosine similarity between u_i and u_j . A shared MLP defines an ordered preference:

$$p_{ij} = g(\rho_{ij}) - g(\rho_{ji}), \quad p_{ij} = -p_{ji}. \quad (8)$$

The relation weights are spectrum-conditioned and normalized over valid non-self candidates. With $w_{ij} \geq 0$, the final score is

$$q_i = c_i + \beta \sum_{j \neq i} w_{ij} p_{ij}. \quad (9)$$

The relation contribution is scattered back to the full candidate pool after coarse top- K_r selection. All operations are shared across candidates and use candidate-attached masks, so jointly permuting a list and its attached features permutes the output scores in the same way, up to tie handling in coarse top- K_r selection. Equation 8 gives raw pairwise antisymmetry independently of the learned parameter values; the weighted contribution in Equation 9 is not itself required to be antisymmetric because w_{ij} and w_{ji} may differ. The model uses no raw peaks, SMILES tokens, precursor mass, formula, or acquisition metadata, and no rank embedding is present in its parameter path.

Structural interpretation. The reversed-pair difference constrains one ordered candidate pair to have a single preference direction: swapping i and j changes only the sign of the raw pair preference p_{ij} . This is a useful inductive bias for local statements of the form “candidate i is preferred to candidate j for this spectrum,” and it is compatible with the pairwise term in the training objective. The shared candidate operations impose the corresponding equivariance at the list level. These are function-class properties, not performance guarantees: pairwise antisymmetry alone does not rule out cycles across three or more candidates, prove better hard-negative discrimination, or imply an advantage over the pointwise control. Those require further controlled experiments.

The implementation exposes an explicit `pair_mode` control. The reported relative model uses `pair_mode=antisymmetric`, which realizes Equation 8; `pair_mode=directed` keeps the same ordered preference network but omits the reverse-pair subtraction as a controlled implementation path, evaluated in Table 4. Older checkpoints without `pair_mode` retain the behavior specified by their legacy `use_antisymmetric` flag.

The near-parameter-matched pointwise control uses the same absolute branch and removes only the relative aggregation. At hidden width 512, the relative model has 1,485,189 trainable parameters and the no-rank pointwise control has 1,478,690, a 0.44% difference. The generic candidate-set Transformer below is retained as a historical reference, not as the proposed mechanism.

4.5 Secondary Historical Reference

For context, we retain the earlier rank-aware candidate-set Transformer as a historical reference. It uses a candidate-attached rank embedding and cross-candidate self-attention [17], unlike the current rank-free relative model. Both historical series use training-only positive forcing, but their model configurations differ; further details are given in Appendix B, but no numerical comparison to the current primary table is made.

4.6 Learning Objective

For the labeled queries $\mathcal{B}_L$ in a minibatch, let V_n be the valid candidate set and y_n the positive index. The primary objective averages listwise cross-entropy over labeled queries:

$$\mathcal{L}_{\text{list}} = -\frac{1}{|\mathcal{B}_L|} \sum_{n \in \mathcal{B}_L} \log \frac{\exp(q_{ny_n})}{\sum_{i \in V_n} \exp(q_{ni})}. \quad (10)$$

We also contrast each positive with every other valid candidate retrieved for that query. The implementation averages over all valid positive-negative pairs in the minibatch:

$$\mathcal{L}_{\text{pair}} = \frac{\sum_{n \in \mathcal{B}_L} \sum_{i \in V_n \setminus \{y_n\}} \log(1 + \exp(q_{ni} - q_{ny_n} + \gamma))}{\sum_{n \in \mathcal{B}_L} (|V_n| - 1)}. \quad (11)$$

The complete objective is

$$\mathcal{L} = \mathcal{L}_{\text{list}} + \lambda_{\text{pair}} \mathcal{L}_{\text{pair}} + \lambda_{\text{spec}} \mathcal{L}_{\text{spec}}. \quad (12)$$

For a cyclicly mismatched spectrum s' from the same minibatch, the spectrum-dependency term contrasts the score of the true candidate under the correct and mismatched query:

$$\mathcal{L}_{\text{spec}} = \text{softplus}(q(s', m^+) - q(s, m^+) + \delta). \quad (13)$$

It is an in-batch dependency constraint, not a claim of causal identification.

The listwise term is a top-one likelihood: it separates the positive from the valid list but does not prescribe a total order among negatives. Both terms are invariant to adding the same query-specific constant to every q_i . Thus, the raw outputs are relative scores, not probabilities; probability calibration is neither performed nor evaluated.

For embedding width d , full-pool coarse scoring costs $O(Kd)$ per query and the relative branch costs $O(K_r^2 d)$ with chunked pair evaluation, rather than quadratic work over all 256 candidates. This is a structural complexity statement. The repository also provides a parameterized forward-only benchmark that varies candidate cutoffs, warm-up and measurement batches, batch size, and device while recording timing metadata; it excludes data loading and does not compute retrieval metrics. The repository retains these scripts as implementation references, but no timing, memory, throughput, or deployment result is claimed in this paper.

5 Experiments

5.1 Dataset and Candidate Pools

We use the published MassSpecGym split [3]: 194,119 training spectra, 19,429 validation spectra, and 17,556 test spectra. These counts total 231,104 spectra; the benchmark reports approximately 29,000 unique molecular structures. Its similarity-aware, structure-disjoint construction reduces close structural overlap between training and test molecules.

MassSpecGym supplies mass- and formula-conditioned candidate settings. The mass pool uses a precursor-mass tolerance; the formula pool assumes a known formula and is not unrestricted annotation. Frozen pools contain at most 256 candidates. The reranker is closed-set: it improves the ordering of supplied candidates but cannot recover a target absent from the pool or generate a new molecule.

Our primary evaluation is the overlap-clean, alignment-42 cache-local matrix, which reuses saved base top-40 lists, embeddings, and local exact-target-SMILES labels. A separate identity audit found agreement with two-dimensional InChIKey labels on these alignment-42 test lists. The official candidate-order audit concerns a separate pre-overlap-clean top-256 pilot; neither audit is fresh official-loader re-encoding or alignment retraining. See Appendix A.

5.2 Implementation Details

The aligned model outputs 512-dimensional embeddings. Its spectrum Transformer has four layers, 16 heads, and width 512; the molecular GINE has four layers, hidden width 128, and dropout 0.2. No published SpecEmbedding checkpoint is loaded, so the stage that would freeze a pretrained spectrum tower is skipped; both towers are optimized end to end on the MassSpecGym training split. Alignment uses AdamW with batch size 128, learning rate and weight decay 10^{-4} , cosine scheduling, gradient-norm clipping at 1, at most 100 epochs, and patience 5. Each epoch makes ten draws per molecular label, randomly selecting an associated spectrum. Spectrum and graph augmentation are independently enabled with probability 0.5. The former removes a floor-rounded 20% of fragment peaks below normalized intensity 0.3 and jitters intensities by $\pm 15\%$; the latter applies 10% node and 10% directed edge-entry dropout. Validation disables augmentation but retains replicate-spectrum sampling under the fixed random stream. The main reranker matrix uses alignment checkpoints produced with seeds 42, 43, and 44. Within each checkpoint, reranker seeds 42–44 control initialization, data-loader order, candidate shuffling, and dropout; they do not turn the alignment checkpoints into independent draws from a known population.

The proposed reranker has hidden width 512, relation width 32, pair chunks of 16, and dropout 0.1. The reported matrix uses $K = 40$ and $K_r = 40$; $K = 256$ is the default full-pool configuration, not the input size of these runs. We use AdamW with learning rate 10^{-4} , batch size 32, at most 30 epochs, and

early stopping on validation MRR with patience 5. The pairwise and spectrum-dependency weights and margins are $\lambda_{\text{pair}} = 0.2$, $\lambda_{\text{spec}} = 0.1$, $\gamma = 0.1$, and $\delta = 0.1$. The near-parameter-matched pointwise control uses the same absolute branch and optimization but no relation aggregation. The legacy candidate-set Transformer is retained only as a historical reference.

Each of the 36 main runs and 30 ablation runs requests at most 20,000 training queries from the historical positive-forced top-40 cache. Validation and test caches remain unforced, and model selection uses validation MRR. These runs are not full-training or all-split no-forcing experiments. Alignment and reranker seeds are recorded separately; their paired combinations describe checkpoint sensitivity, not population-level uncertainty.

5.3 Code, Data, and Artifact Availability

The anonymous supplement provides source inspection, locked environment files, automated tests, and a tiny synthetic CPU smoke test. It does not provide the MassSpecGym spectra, candidate pools, trained checkpoints, or caches. The repository’s reproducibility index lists the local checkpoints, resolved configurations, logs, hashes, and selected ranking outputs used for the reported results. The real-data entry point is `run_rerank_pipeline.py`; additional implementation and reproducibility notes are given in Appendix D. The repository source also contains a query-level case exporter that records candidate identities and base/reranker ranks for selected cache queries. It provides ranking evidence for subsequent manual inspection, not fragment attribution or a chemical causal interpretation.

5.4 Baselines and Metrics

The controlled comparisons are the aligned cosine retriever, the near-parameter-matched no-rank pointwise scorer, and the relative reranker. The earlier candidate-set Transformer is retained only as a historical reference. We also run a local saved-embedding cosine control that matches the scoring form associated with JESTR; it is a reimplemented control, not an official JESTR encoder/checkpoint reproduction. Because both this control and the aligned base retriever use cosine scores on the same saved normalized embeddings, their local numbers coincide; this is a protocol-matched scoring control rather than an independent JESTR comparison. GLMR remains reported-only because its conditional generation pipeline is not reproduced here.

We report Recall@1, Recall@5, Recall@10, Recall@20, and mean reciprocal rank (MRR) over all 17,556 test queries. The primary table is a cache-local, evaluation at overlap-clean alignment seed 42, using saved top-40 candidate order and exact-target-SMILES labels, without test-time forcing. It uses three reranker training seeds (42–44) and at most 20,000 training queries. A separate table reports the same pointwise/relative comparison for alignment seeds 42–44. A tracked official-compatible candidate-order and identity audit is reported as a protocol check, not as fresh loader re-encoding. Means and sample SDs within an

Table 2. Primary cache-local top-40 evaluation at the overlap-clean alignment-42 checkpoint. Values are percentages; learned rows are means $\pm$ sample SD over reranker training seeds 42–44. All methods rank the same saved test pool and exact-target-SMILES labels, with no test forcing. Training uses positive-forced caches and a 20,000-query cap. The expanded alignment sensitivity is reported in Table 3.

Pool	Method	R@1	R@5	R@10	R@20	MRR
Mass	SpecMolAlign (saved top-40 pool)	47.46	63.51	71.02	77.76	55.03
	Pointwise control	58.12 $\pm$ 0.54	73.61 $\pm$ 0.54	78.63 $\pm$ 0.48	82.24 $\pm$ 0.10	64.91 $\pm$ 0.47
	Relative candidate model	57.88 $\pm$ 0.59	73.55 $\pm$ 0.58	78.72 $\pm$ 0.29	82.15 $\pm$ 0.13	64.78 $\pm$ 0.52
Formula	SpecMolAlign (saved top-40 pool)	63.10	75.79	80.87	84.93	68.86
	Pointwise control	70.76 $\pm$ 0.59	80.07 $\pm$ 0.35	84.33 $\pm$ 0.08	86.86 $\pm$ 0.08	75.17 $\pm$ 0.41
	Relative candidate model	70.88 $\pm$ 0.38	80.60 $\pm$ 0.37	84.68 $\pm$ 0.26	86.85 $\pm$ 0.09	75.38 $\pm$ 0.27

Table 3. Descriptive alignment-checkpoint sensitivity under the expanded top-40 protocol: training forcing, unforced validation/test. Learned values are means over reranker seeds 42–44; the cache-local matrix is not a fresh official-loader re-encoding.

Alignment	Pool	Base R@1/MRR	Pointwise R@1/MRR	Relative R@1/MRR
42	Mass	47.46/.5503	58.12/.6491	57.88/.6478
43	Mass	35.92/.4433	58.32/.6398	57.61/.6350
44	Mass	23.91/.3226	46.42/.5238	47.05/.5284
42	Formula	63.10/.6886	70.76/.7517	70.88/.7538
43	Formula	45.08/.5185	64.72/.6836	64.10/.6795
44	Formula	28.73/.3687	54.26/.5877	54.35/.5911

alignment describe reranker dispersion; the cross-alignment rows are descriptive checkpoint sensitivity, not independent population replicates. The current paper does not claim official JESTR reproduction, statistical significance, or end-to-end deployment performance.

5.5 Main Results

Both learned models improve on their alignment-42 base in both pools. Pointwise is slightly higher than relative at R@1 and MRR for mass (58.12 vs. 57.88 and 0.6491 vs. 0.6478), whereas relative is slightly higher for formula (70.88 vs. 70.76 and 0.7538 vs. 0.7517). These descriptive differences do not establish an independent relative-module effect. Evaluation-only candidate-cutoff files are retained in the repository; they are not separately trained models and are not used as an additional primary result.

This controlled comparison gives the paper a two-part result. Supervised second-stage scoring is useful within the supplied candidate pool, but adding the relative interaction is not required for the observed improvement and is not uniformly better than the matched pointwise scorer. The latter result is a boundary of the proposed mechanism, not a failed comparison to be hidden.

Across nine alignment–reranker–seed pairs per candidate protocol, relative minus pointwise has mean R@1 differences of -0.11 and -0.14 percentage points for mass and formula, respectively; the corresponding mean MRR differences are -0.0005 and $+0.0005$. Relative wins five of nine R@1 pairs in each candidate

Table 4. Relative-module ablations under the cache-local top-40 protocol, with training forcing and unforced validation/test. R@1/MRR use percentages and decimals, respectively; each entry is a mean $\pm$ sample SD over reranker seeds 42–44.

Variant	Mass R@1/MRR		Formula R@1/MRR	
Full antisymmetric	57.88 $\pm$ 0.59	/.6478 $\pm$.0052	70.88 $\pm$ 0.38	/.7538 $\pm$.0027
Directed pair	57.97 $\pm$ 0.49	/.6471 $\pm$.0047	70.06 $\pm$ 1.29	/.7471 $\pm$.0106
No spectrum conditioning	58.42 $\pm$ 0.81	/.6522 $\pm$.0069	71.01 $\pm$ 0.40	/.7546 $\pm$.0033
No $\mathcal{L}_{\text{pair}}$	58.01 $\pm$ 0.38	/.6488 $\pm$.0037	70.91 $\pm$ 0.36	/.7541 $\pm$.0027
No $\mathcal{L}_{\text{spec}}$	58.44 $\pm$ 0.54	/.6516 $\pm$.0064	70.59 $\pm$ 0.79	/.7515 $\pm$.0056

protocol. These are descriptive paired estimates, not significance tests, and they support the boundary claim that supervised reranking is more stable than any independent relative-module advantage.

Mechanism analysis. Table 4 reports the complete 30-run relative-module ablation matrix: five relative configurations, two candidate protocols, and three reranker seeds at the alignment-42 checkpoint. The full configuration is the antisymmetric, spectrum-conditioned model. Removing spectrum conditioning raises mean R@1 and MRR in both pools; other changes have metric- and pool-dependent directions. These aggregate results do not establish a benefit for spectrum conditioning or a causal ranking of the components.

JESTR and GLMR remain reported-only, not reproduced, and not directly comparable background methods. We retain their mechanism descriptions but do not reproduce their numeric table or infer a relative ranking.

Difficulty strata. In the seed-42 local query analysis, the selected labeled subsets contain 14,726 mass and 15,670 formula queries. For the relative model, R@1 gains over the base at base ranks 2–5/6–20/21–40 are +54.05/+38.13/+29.98 percentage points for mass and +55.41/+34.18/+19.10 for formula. Base-rank-1 queries instead decline to 87.41% (mass) and 95.12% (formula), from 100% under the base stratum. The pointwise control shows the same unresolved-query pattern; these strata are descriptive and do not establish a causal relation-module effect.

Historical baseline boundary. Earlier rank-aware Transformer results also used top-40 caches and training-only positive replacement. They are retained only as historical context, not as measurements of the present rank-free model or evidence for the relative module introduced here; further details appear in Appendix B.

6 Discussion

The expanded cache-local matrix separates candidate coverage from list ordering. At alignment-42, the saved top-40 test lists give mass and formula upper bounds of 83.88% and 89.26%, respectively. Across three alignment checkpoints

and three reranker seeds per checkpoint, both learned variants improve their corresponding base rows. The relative-pointwise ordering varies by pool, metric and checkpoint. The result supports supervised second-stage reranking as a framework component, but does not isolate an independent relative-module benefit.

The 30-run ablation matrix shows that directed versus antisymmetric pairs, spectrum conditioning, and the two auxiliary losses affect the candidate pools and metrics differently. These observations are descriptive: they clarify the behavior of the proposed module, but do not establish a causal component gain.

The relative branch also has a higher structural cost than pointwise scoring: the full-pool absolute branch is $O(Kd)$, while pair evaluation on the coarse subset is $O(K_r^2d)$. The repository retains fixed-hardware forward-only measurements as implementation references; they are not end-to-end latency, throughput, or deployment claims in this paper.

The primary matrix reuses a cache-local candidate order and saved alignment embeddings. A separate official-compatible audit uses the published candidate JSON and the pilot’s saved embeddings, but does not re-encode spectra/molecules through a fresh official loader or retrain the first stage. External methods and other candidate constructions are outside the comparison.

7 Limitations and Responsible Evaluation

The expanded reranker matrix requests up to 20,000 training queries, evaluates three alignment checkpoints, and repeats each principal configuration with three reranker seeds. It uses top-40 training caches with positive forcing; validation/test are unforced. It therefore does not establish the effect of full-training, top-256, all-split no-forcing operation. This provides descriptive checkpoint sensitivity, not a confidence interval, significance test, or complete end-to-end seed analysis. The evaluation also reuses saved embeddings and supplied candidate pools; it does not claim fresh official-loader re-encoding, external-baseline reproduction, or candidate-aware alignment retraining. These limitations bound the scope of the two-part result: the framework-level reranking observation is supported within the evaluated checkpoints, whereas the relative module’s generality remains open.

The method is closed-set and formula-conditioned when the formula candidate protocol is used. It cannot recover a target absent from the candidate pool, generate a molecule outside that pool, or generalize automatically to other candidate constructions and acquisition conditions. The spectrum tower uses selected peaks and precursor m/z but no explicit adduct, instrument, or collision-energy metadata. Its pairwise structural constraints likewise do not provide probability calibration or a guarantee of globally consistent ranking. Further protocol and reproducibility notes appear in Appendices A and D.

8 Conclusion

We introduced SpecEmbedding-Rerank, a rank-free second-stage reranker that combines full-pool coarse scoring with spectrum-conditioned relative candidate preferences. Across three alignment checkpoints with three reranker seeds per checkpoint, the expanded cache-local matrix showed that both the relative model and its near-parameter-matched pointwise control improved their corresponding base in mass- and formula-conditioned retrieval. The primary table uses the overlap-clean alignment-42 top-40 cache and models trained with a 20,000-query cap and training-only positive forcing. Separate audits check candidate order and identity within their respective saved-cache scopes. The relative-pointwise ordering depends on the pool and metric, and the 30-run ablation matrix did not establish consistent benefits for the tested components. The evidence therefore supports a carefully scoped supervised reranking framework and makes the relative interaction’s boundary explicit: it is a principled candidate- relation mechanism, but the current evidence does not establish an independent or generally robust benefit over pointwise scoring. The method remains limited to the fixed saved-embedding candidate protocol.

A Evaluation Protocol and Candidate Identity

The primary matrix uses local exact-target-SMILES labels and saved base top-40 lists. On the overlap-clean alignment-42 test caches, a separate two-dimensional InChIKey audit found no extra positives or candidate-identity collisions: 14,726 mass and 15,670 formula queries have a positive among 17,556 queries per pool. Thus the tested identity rules agree on those lists; missing positives remain coverage misses. This audit is not automatically transferable to other alignment checkpoints or candidate cutoffs.

The official-order audit instead uses a separate pre-overlap-clean alignment and top-256 no-forcing pilot (5,000 training queries, three epochs, reranker seeds 42–44). Its candidate sets match the MassSpecGym 1.3.1 retrieval JSON; a full-pool identity check found one equivalent positive per test query. Reordering those lists and rescored saved embeddings does not rerun official loader transforms or retrain alignment, and does not validate the main top-40 matrix as an official evaluation.

The spectrum-record audit found six validation and three test records matching training inputs with different molecular labels. The affected validation records are excluded from overlap-clean alignment validation; this does not imply that every later reranker run applies the same exclusions. Candidate-cutoff analyses are evaluation-only: the pilot truncates top-256 caches, while the expanded matrix truncates its own top-40 caches. Neither is cutoff-specific retraining.

B Implementation and Historical Baselines

The current training entry point writes `data_summary` into new reranker checkpoints: actual labeled training-query count, requested cap, `train_k`, model type, pair mode, seed, and available cache metadata. These fields distinguish training scope and forcing semantics; they do not change the forward path or substitute for experiments.

The historical Candidate-Set Transformer uses rank embeddings and generic candidate self-attention. It shares the historical training-forcing convention with the reported top-40 matrix, but differs in architecture and capacity. It is not the near-parameter-matched pointwise control. The default top-256 no-forcing configuration and these historical measurements remain separate.

C Additional Mechanism Analyses

The 30-run matrix in Table 4 contains five configurations across two candidate protocols and three reranker seeds. The difficulty strata in the main text describe where the saved pool leaves ranking headroom; they do not identify a causal relation-module effect. Evaluation cutoffs and fixed-hardware forward measurements are implementation references, not retraining or end-to-end deployment benchmarks.

D Reproducibility and Responsible-Evaluation Notes

The reproducibility index records source commits, resolved configurations, cache metadata, checkpoint/log hashes and selected ranking outputs for the reported matrix. The anonymous supplement provides source inspection, locked environment files, tests and a synthetic CPU smoke test. It excludes real spectra, candidate pools, caches, checkpoints and internal artifact inventories; the smoke therefore validates execution, not the scientific metrics.

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